

A proof of the Apéry Hankel growth conjecture

OEIS A228143, linearly shifted determinants,
and a non-D-finite generating series

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Abstract

For the Apéry numbers $A_k = \sum_{j=0}^k \binom{k}{j}^2 \binom{k+j}{j}^2$, let $D_n = \det(A_{i+j})_{0 \leq i, j \leq n}$. The OEIS entry A228143 records a conjecture, added by Vaclav Kotesovec on 13 September 2026, that $D_n^{1/n^2} \rightarrow (1 + \sqrt{2})^4/4$. Starting from Edgar's known positive moment-density theorem, we prove this limit and the stronger estimate $\log D_n = n(n+1) \log((1 + \sqrt{2})^4/4) + O(n)$. The proof uses a polynomial lower bound for the density, monic orthogonal polynomials, and elementary Legendre–Chebyshev estimates. We also obtain an explicit growth law for $\det(A_{m(i+j)+r_n})$ when m is fixed and r_n/n converges, including nonzero linear shifts. A further consequence is that the formal ordinary generating series of D_n is not D-finite. The package includes exact determinants through $n = 200$, independent arithmetic checks, and runnable Python and C++/GMP code. The density representation is credited, not reproved; the subsequent determinant arguments are given in full.

Main result

$$\lim_{n \rightarrow \infty} D_n^{1/n^2} = \frac{17 + 12\sqrt{2}}{4} = 8.4926406871192851464 \dots$$

More precisely, some $\kappa > 0$ satisfies, for every $n \geq 0$,

$$\kappa^n \left(\frac{17 + 12\sqrt{2}}{4} \right)^{n(n+1)} \leq D_n \leq 4^n \left(\frac{17 + 12\sqrt{2}}{4} \right)^{n(n+1)}.$$

Status and attribution. This is a proof of the precise assertion still labelled a conjecture in the accessed OEIS revision, not a claim that the result has no antecedent anywhere in the literature. The essential pre-existing analytic input is G. A. Edgar's work. This report has not been peer reviewed or formalized in a proof assistant.

1 The selected problem and the stronger result

The Apéry numbers, OEIS A005259, are

$$A_k = \sum_{j=0}^k \binom{k}{j}^2 \binom{k+j}{j}^2, \quad A_0 = 1, \quad A_1 = 5, \quad A_2 = 73. \quad (1.1)$$

Define the leading Hankel determinants

$$D_n = \det(A_{i+j})_{0 \leq i, j \leq n}. \quad (1.2)$$

Thus the matrix has order $n + 1$, not n . The first values are 1, 48, 161856, 39002646528, ...; they agree with OEIS A228143 [2, 1].

The target is the asymptotic formula in the %F field of revision #40 of A228143, dated 13 September 2026. It is attributed to Vaclav Kotesovec and asserts

$$\lim_{n \rightarrow \infty} D_n^{1/n^2} = \frac{(1 + \sqrt{2})^4}{4}. \quad (1.3)$$

It is distinct from the divisibility and root-integrality assertions also recorded in that entry. The source status here refers to the version accessed on 18 September 2026 [1].

Use the constants

$$C = 17 + 12\sqrt{2} = (1 + \sqrt{2})^4, \quad c = C^{-1} = 17 - 12\sqrt{2}, \quad \Lambda = \frac{C}{4}. \quad (1.4)$$

The small positive number c is an interior singular point of the moment density; it is *not* the left endpoint of its support.

Theorem 1.1. *There exists $\kappa \in (0, 1]$ such that*

$$\kappa^n \Lambda^{n(n+1)} \leq D_n \leq 4^n \Lambda^{n(n+1)} \quad (n \geq 0). \quad (1.5)$$

Consequently,

$$\log D_n = n(n+1) \log \Lambda + O(n), \quad (1.6)$$

$$D_n^{1/n^2} = \Lambda \exp(O(1/n)), \quad (1.7)$$

$$\left(\frac{D_n}{D_{n-1}} \right)^{1/(2n)} \rightarrow \Lambda \quad (n \rightarrow \infty). \quad (1.8)$$

In particular, (1.3) is true.

The proof will bound each monic orthogonal-polynomial norm separately. This is stronger than trying to estimate one large determinant after replacing its entries by asymptotic approximations. The latter operation is generally unsafe; Section 6 gives an exact counterexample to that method.

2 The known moment density and a global lower bound

Known input. Edgar's Theorem 1, Proposition 4, Notation 6, Propositions 25–26, and Corollary 27 [3] give an integrable density φ on $[0, C]$ such that

$$A_k = \int_0^C x^k \varphi(x) dx, \quad \int_0^C \varphi(x) dx = 1. \quad (2.1)$$

It is positive and real analytic on $(0, c)$ and (c, C) , with

$$\varphi(x) = -\frac{6}{\pi^2} \log x + o(1) \quad (x \downarrow 0), \quad (2.2)$$

$$\varphi(x) = \frac{(C-x)^{1/2}}{2^{5/4}C\pi^2} + O((C-x)^{3/2}) \quad (x \uparrow C). \quad (2.3)$$

At c , on each side, it is a nonzero combination of convergent Frobenius solutions with leading orders $0, \frac{1}{2}, 1$. Here our C, c correspond to Edgar's c, c_0 .

In particular, for positive distance t from c on either side, the first nonzero term gives

$$\varphi(c \pm t) = a_{\pm} t^{\alpha_{\pm}} (1 + o(1)), \quad a_{\pm} > 0, \quad \alpha_{\pm} \in \{0, \frac{1}{2}, 1\}. \quad (2.4)$$

Indeed, the three basis solutions start with distinct powers. A nonzero solution has a first nonzero coefficient, and positivity fixes the sign of its leading term. No value of $\varphi(c)$ needs to be assigned, and equality of its one-sided limits is not required below.

Lemma 2.1 (Polynomial minorant). *Let $Q(x) = (x - c)(x - C)$. There is $\eta > 0$ such that*

$$\varphi(x) \geq \eta Q(x)^2 \quad \text{for almost every } x \in [0, C]. \quad (2.5)$$

Proof. Consider $R(x) = \varphi(x)/Q(x)^2$ away from c, C . On each of $(0, c)$ and (c, C) this is continuous and positive. Since $cC = 1$, $Q(0) = 1$, and (2.2) implies $R(x) \rightarrow +\infty$ as $x \downarrow 0$. Equation (2.3) gives $R(x) \asymp (C-x)^{-3/2} \rightarrow +\infty$ as $x \uparrow C$. At c , (2.4) gives $R(c \pm t) \asymp t^{\alpha_{\pm}-2} \rightarrow +\infty$. Choose small neighborhoods of these three points on which R is bounded below by a positive constant. On the remaining compact pieces, its positive continuous minimum is also positive. The least of these bounds is an admissible η . $\square$

The interior singularity is the delicate point in this argument. Positivity alone would not justify a strictly positive uniform lower bound for the density. Squaring a polynomial that vanishes at c and C avoids that incorrect inference. The constant η is established to exist; this report does not claim a certified numerical value for it.

3 Elementary norm estimates and the proof

3.1 A moment determinant is a product of minimum norms

For a nonzero real polynomial p , positivity of φ on an interval gives $\int p(x)^2 \varphi(x) dx > 0$. Thus its polynomial inner product is positive definite. Let p_k be its monic orthogonal polynomial of degree k , and put

$$h_k = \int_0^C p_k(x)^2 \varphi(x) dx = \min_{\substack{p \in \mathbb{R}[x] \text{ monic} \\ \deg p = k}} \int_0^C p(x)^2 \varphi(x) dx. \quad (3.1)$$

The minimum follows by writing $p = p_k + q$, $\deg q < k$, and using orthogonality. The change of basis from $1, x, \dots, x^n$ to $p_0, \dots, p_n$ is unitriangular. Its determinant is one, so taking Gram determinants gives

$$D_n = \prod_{k=0}^n h_k, \quad h_0 = 1. \quad (3.2)$$

These are the usual monic-norm identities; compare DLMF §18.2 [4]. The argument above also proves $D_n > 0$ for every n .

3.2 The Lebesgue lower comparison

Lemma 3.1. *For $L > 0$ and $k \geq 0$, the minimum squared norm of a monic polynomial of degree k for Lebesgue measure on $[0, L]$ is*

$$\ell_k(0; L) = \frac{L^{2k+1}}{(2k+1)\binom{2k}{k}^2} \geq L \left(\frac{L}{4}\right)^{2k}. \quad (3.3)$$

Proof. On $[0, 1]$, the polynomial

$$R_k(t) = \frac{1}{k!} \frac{d^k}{dt^k} [t^k(t-1)^k]$$

is orthogonal to all lower-degree polynomials by k integrations by parts. Its leading coefficient is $\binom{2k}{k}$. Another integration by parts gives

$$\int_0^1 R_k(t) t^k dt = \int_0^1 t^k(1-t)^k dt = \frac{(k!)^2}{(2k+1)!}.$$

Orthogonality therefore implies $\int_0^1 R_k(t)^2 dt = 1/(2k+1)$. Dividing by its leading coefficient and scaling $x = Lt$ proves the equality in (3.3) and the minimizing property.

For the inequality, set $u_k = (2k+1)\binom{2k}{k}^2/16^k$. Direct cancellation gives $u_0 = 1$ and

$$\frac{u_{k+1}}{u_k} = 1 - \frac{1}{4(k+1)^2} < 1.$$

Thus $(2k+1)\binom{2k}{k}^2 \leq 16^k$, as needed. □

If p is monic of degree k , then pQ is monic of degree $k+2$. Lemmas 2.1 and 3.1 give

$$\int_0^C p(x)^2 \varphi(x) dx \geq \eta \int_0^C (p(x)Q(x))^2 dx \geq \eta \ell_{k+2}(0; C) \geq \eta C \Lambda^{2k+4}.$$

Consequently, with

$$\kappa = \eta C \Lambda^4 > 0, \quad h_k \geq \kappa \Lambda^{2k} \quad (k \geq 0). \quad (3.4)$$

The case $k = 0$, together with $h_0 = 1$, shows $\kappa \leq 1$.

3.3 The Chebyshev upper comparison

The Chebyshev polynomial T_k satisfies $T_k(\cos \theta) = \cos(k\theta)$. The recurrence $T_{k+1}(t) = 2tT_k(t) - T_{k-1}(t)$ shows that its leading coefficient is 2^{k-1} for $k \geq 1$. Hence

$$q_k(x) = 2\Lambda^k T_k\left(\frac{2x}{C} - 1\right)$$

is monic of degree k and has absolute value at most $2\Lambda^k$ on $[0, C]$. Since the measure has total mass one, the minimum in (3.1) satisfies

$$h_k \leq \int_0^C q_k(x)^2 \varphi(x) dx \leq 4\Lambda^{2k} \quad (k \geq 1). \quad (3.5)$$

Proof of Theorem 1.1. Multiply (3.4) and (3.5) over $k = 1, \dots, n$ and use (3.2) with $h_0 = 1$. Since $2 \sum_{k=1}^n k = n(n+1)$, this is (1.5). Taking logarithms gives (1.6); dividing by n^2 gives (1.7). Finally $D_n/D_{n-1} = h_n$, and the individual bounds imply

$$\Lambda \kappa^{1/(2n)} \leq h_n^{1/(2n)} \leq \Lambda 4^{1/(2n)},$$

which proves (1.8). $\square$

Remark 3.2. The proof does not assert that h_n/Λ^{2n} converges. It proves this sequence is bounded above and bounded away from zero. Nor does (1.6) give a full multiplicative asymptotic equivalent for D_n .

4 An explicit law for linearly shifted determinants

The same comparison idea gives more than fixed shifts. For integers $m \geq 1$ and $r \geq 0$, define

$$H_n^{(m,r)} = \det(A_{m(i+j)+r})_{0 \leq i,j \leq n}. \quad (4.1)$$

Here m is the stride in the moment indices. The shift r is held constant throughout each matrix, but may vary from one matrix size to another.

Define, for $\sigma \geq 0$,

$$\Psi(\sigma) = \frac{1}{2} [(\sigma+2)^2 \log(\sigma+2) - 2(\sigma+1)^2 \log(\sigma+1) + \sigma^2 \log \sigma], \quad (4.2)$$

with $0^2 \log 0 := 0$.

Theorem 4.1 (Linear-shift growth law). *Fix a positive integer m . If r_n are nonnegative integers and $r_n/n \rightarrow \rho \in [0, \infty)$, then*

$$\lim_{n \rightarrow \infty} \frac{\log H_n^{(m,r_n)}}{n^2} = (m + \rho) \log C - \Psi(\rho/m). \quad (4.3)$$

Equivalently,

$$\lim_{n \rightarrow \infty} (H_n^{(m,r_n)})^{1/n^2} = C^{m+\rho} \exp(-\Psi(\rho/m)).$$

In particular, every sublinear shift $r_n = o(n)$ has limit $C^m/4$.

For example,

$$\lim_{n \rightarrow \infty} \det(A_{2i+2j})_{0 \leq i,j \leq n}^{1/n^2} = \frac{C^2}{4}, \quad (4.4)$$

$$\lim_{n \rightarrow \infty} \det(A_{i+j+n})_{0 \leq i,j \leq n}^{1/n^2} = \frac{16C^2}{81\sqrt{3}}. \quad (4.5)$$

The constant $C^m/4$ for a stride m is not $(C/4)^m$. The proof below also provides a finite-size comparison with a completely explicit determinant.

4.1 The power-weight model

For $s > -1$ and $L > 0$, put

$$\mathcal{J}_n(s; L) = \det \left(\frac{L^{i+j+s+1}}{i+j+s+1} \right)_{0 \leq i,j \leq n}. \quad (4.6)$$

This is the moment determinant of $y^s dy$ on $[0, L]$. Its monic minimum norms are

$$\ell_k(s; L) = \frac{L^{2k+s+1}}{(2k+s+1) \binom{2k+s}{k}^2}, \quad \binom{2k+s}{k} = \prod_{j=1}^k \frac{k+s+j}{j}. \quad (4.7)$$

To verify (4.7), replace the Rodrigues polynomial in Lemma 3.1 by

$$R_{k,s}(t) = \frac{t^{-s}}{k!} \frac{d^k}{dt^k} [t^{k+s}(t-1)^k].$$

It is a polynomial of degree k , with leading coefficient $\binom{2k+s}{k}$. Integration by parts proves orthogonality for weight t^s . The boundary terms vanish because $s > -1$. Its pairing with t^k is the beta integral $\int_0^1 t^{k+s}(1-t)^k dt$; dividing by the leading coefficient and scaling the interval gives (4.7).

We will use both its norm product and its Cauchy determinant form:

$$\mathcal{J}_n(s; L) = \prod_{k=0}^n \ell_k(s; L) = L^{(n+1)(n+s+1)} \frac{\prod_{j=0}^n (j!)^2}{\prod_{i=0}^n \prod_{j=0}^n (s+1+i+j)}. \quad (4.8)$$

For completeness, the second equality follows by factoring powers of L and applying the Cauchy identity

$$\det \left(\frac{1}{x_i + y_j} \right) = \frac{\prod_{i < j} (x_j - x_i)(y_j - y_i)}{\prod_{i,j} (x_i + y_j)}$$

with $x_i = i$, $y_j = s+1+j$. One elementary proof of that identity clears the common denominator: the resulting polynomial is alternating in the x_i and in the y_j , so is divisible by both Vandermonde products. Total degrees agree. The remaining constant is one, by induction on the residue at $x_n = -y_n$, starting with the 1×1 case.

4.2 Uniform comparison with the power-weight model

Proposition 4.2. *For fixed $m \geq 1$ and $T < \infty$, uniformly for integers $0 \leq r \leq Tn$,*

$$\log H_n^{(m,r)} = \log \mathcal{J}_n(r/m; C^m) + O_{m,T}(n) \quad (n \geq 1). \quad (4.9)$$

Proof. Set $L = C^m$, $a = c^m$, $s = r/m$, and change variables $y = x^m$ in (2.1). The moments in (4.1) are those of $y^s w_m(y) dy$, where

$$w_m(y) = \frac{1}{m} y^{1/m-1} \varphi(y^{1/m}), \quad 0 < y < L. \quad (4.10)$$

The map is smooth with nonzero derivative near a and L . Thus the one-sided finite orders at a and the square-root order at L are unchanged. At zero,

$$w_m(y) \sim \frac{6}{m^2 \pi^2} y^{1/m-1} \log(1/y) \rightarrow +\infty.$$

The argument of Lemma 2.1 therefore supplies $\eta_m > 0$ with $w_m(y) \geq \eta_m [(y-a)(y-L)]^2$ almost everywhere.

Set $\beta = -1 + 1/(2m) \in (-1, 0)$. Near zero, $w_m(y)/y^\beta \rightarrow 0$, and this ratio is bounded on the remaining intervals, including their one-sided limits at a and L . Hence a constant $K_m > 0$ also exists such that

$$\eta_m [(y-a)(y-L)]^2 \leq w_m(y) \leq K_m y^\beta. \quad (4.11)$$

Neither constant depends on r or n .

Let $h_k(s)$ denote the monic minimum norm for $y^s w_m(y) dy$. Multiplication by the monic quadratic on the left of (4.11), and a minimizing trial polynomial for the upper power weight, give

$$\eta_m \ell_{k+2}(s; L) \leq h_k(s) \leq K_m \ell_k(s + \beta; L).$$

The parameter $s + \beta$ is greater than -1 . Multiplying over $k = 0, \dots, n$ yields

$$\eta_m^{n+1} \mathcal{J}_n(s; L) \frac{\ell_{n+1}(s; L) \ell_{n+2}(s; L)}{\ell_0(s; L) \ell_1(s; L)} \leq H_n^{(m,r)} \leq K_m^{n+1} \mathcal{J}_n(s + \beta; L). \quad (4.12)$$

We check that both corrections to $\log \mathcal{J}_n(s; L)$ are $O(n)$. Write $b = -\beta \in (0, 1)$. Formula (4.8) gives

$$\log \frac{\mathcal{J}_n(s + \beta; L)}{\mathcal{J}_n(s; L)} = (n+1)\beta \log L - \sum_{i,j=0}^n \log \left(1 - \frac{b}{s+1+i+j} \right).$$

For $t \geq 1$, $0 \leq -\log(1 - b/t) \leq b/((1-b)t)$. Counting pairs with fixed $i + j$ gives

$$\sum_{i,j=0}^n \frac{1}{s+1+i+j} \leq \sum_{i,j=0}^n \frac{1}{1+i+j} \leq 2n+1.$$

Thus the logarithmic upper correction is $O_m(n)$, uniformly in $s \geq 0$.

For the lower correction, $s \leq (T/m)n$. At $k = 0, 1, n+1, n+2$, formula (4.7) shows $|\log \ell_k(s; L)| = O_{m,T}(n)$. For $k = 0, 1$ this is immediate. For the other two indices use

$$1 \leq \binom{2k+s}{k} \leq \frac{(2k+s)^k}{k!} \leq \left(\frac{e(2k+s)}{k} \right)^k,$$

whose logarithm is $O_{m,T}(n)$ since $k \geq n+1$ and $s = O(n)$. Here $k! \geq (k/e)^k$ follows by integrating $\log x$ on $[1, k]$. The remaining factors in (4.7) have logarithms $O(n)$ as well. Applying these estimates to (4.12) proves (4.9). $\square$

4.3 Evaluating the limiting constant

Lemma 4.3. *If $s_n \geq 0$ and $s_n/n \rightarrow \sigma \geq 0$, then*

$$\frac{1}{n^2} \log \mathcal{J}_n(s_n; L) \rightarrow (1 + \sigma) \log L - \Psi(\sigma). \quad (4.13)$$

Proof. Taking logarithms in (4.8), the factorial part satisfies

$$2 \sum_{j=0}^n \log(j!) = n^2 \log n - \frac{3}{2} n^2 + O(n \log(n+1)). \quad (4.14)$$

For example, interchange the sums to obtain $2 \sum_{k=1}^n (n+1-k) \log k$ and compare the sums of $\log k$ and $k \log k$ with their integrals. For the denominator, Riemann sums give

$$\frac{1}{n^2} \sum_{i,j=0}^n \log(s_n + 1 + i + j) - \log n \rightarrow \int_0^1 \int_0^1 \log(\sigma + x + y) dx dy. \quad (4.15)$$

When $\sigma > 0$ the integrand is continuous. When $\sigma = 0$, split off $i + j \leq \delta n$. The normalized absolute contribution there is bounded by $O(\delta^2(1 + |\log \delta|)) + o(1)$: count the $i + j = k$ pairs and use $\log(n/(k+1))$ as an upper bound for the absolute logarithm for large n . Outside this corner, convergence is uniform. Letting $\delta \downarrow 0$ justifies (4.15) also at $\sigma = 0$.

Two integrations, with $t^2 \log t$ assigned its continuous value at zero, yield

$$\int_0^1 \int_0^1 \log(\sigma + x + y) dx dy = \Psi(\sigma) - \frac{3}{2}.$$

The two $\frac{3}{2}$ terms cancel between this expression and (4.14). The power of L in (4.8) contributes $(1 + \sigma) \log L$, proving the claim. $\square$

Proof of Theorem 4.1. The convergence $r_n/n \rightarrow \rho$ implies $r_n \leq Tn$ eventually for some finite T . Apply Proposition 4.2, then Lemma 4.3 with $s_n = r_n/m$, $L = C^m$, and $\sigma = \rho/m$. The error $O(n)$ disappears after division by n^2 . Finally $\Psi(0) = \log 4$. $\square$

For fixed m, r , (4.9) and the norm formula also give $\log H_n^{(m,r)} = n(n+1) \log(C^m/4) + O_{m,r}(n)$. To see this without a further asymptotic theorem, changing the power parameter from 0 to a fixed $s = r/m$ changes $\log J_n$ by $O_s(n)$, using $\log(1 + s/t) \leq s/t$ and the same reciprocal sum estimate as above. Lemma 3.1 and the Chebyshev upper comparison for Lebesgue measure then bound $\log J_n(0; L) - n(n+1) \log(L/4)$ by $O_L(n)$. The stride m is fixed throughout these statements; no uniform assertion for $m \rightarrow \infty$ is made.

5 A generating-function consequence

Theorem 5.1. *The sequence (D_n) is not P-recursive over $\mathbb{C}$. Its formal ordinary generating series $F(z) = \sum_{n \geq 0} D_n z^n$ is not D-finite. Moreover, for every fixed real s , the series $\sum_{n \geq 0} D_n z^n / (n!)^s$ has radius of convergence zero.*

Proof. Suppose that, for all sufficiently large integers n ,

$$\sum_{j=0}^q P_j(n) D_{n+j} = 0, \quad P_j \in \mathbb{C}[n], \quad P_q \neq 0. \quad (5.1)$$

The case $q = 0$ contradicts $D_n > 0$. For $q \geq 1$, the leading coefficient polynomial has no zeros at sufficiently large integers. Ratios of the other coefficient polynomials to it have at most polynomial growth. Therefore some $K > 0$, integer $d \geq 0$, and N satisfy

$$|D_{n+q}| \leq K n^d \max_{0 \leq j < q} |D_{n+j}| \quad (n \geq N).$$

Taking running maxima and increasing K if necessary gives $|D_n| \leq K_0 K_1^n (n!)^d$. Thus $\log D_n = O(n \log n)$, contradicting (1.6), since $\Lambda > 1$. This proves the first assertion.

A polynomial-coefficient linear differential equation for F would, by coefficient comparison, give a nontrivial polynomial-coefficient recurrence for its coefficients for all sufficiently large indices. Hence F cannot be D-finite. This argument is about formal power series and does not assume that F converges.

Finally,

$$\frac{1}{n} \log \frac{D_n}{(n!)^s} = n \log \Lambda + O(\log(n+1)) \rightarrow +\infty.$$

The root test gives radius zero for every fixed s . $\square$

Divergence alone would *not* have proved non-D-finiteness: a formal D-finite series can have radius zero. The polynomial-recurrence growth bound is the essential extra argument.

6 Why entrywise asymptotics do not determine this limit

Consider, for $0 \leq a < C$, the elementary moments

$$m_k^{(a)} = \int_a^C x^k dx = \frac{C^{k+1} - a^{k+1}}{k+1}.$$

For each fixed $0 < a < C$,

$$\frac{m_k^{(a)}}{m_k^{(0)}} = 1 - (a/C)^{k+1}.$$

Thus these two moment sequences agree to every algebraic relative asymptotic order in $1/k$, with an exponentially small relative difference. Nevertheless their determinant limits differ. Translation by a is a unitriangular change of the monomial basis, so it does not change a Gram determinant. Applying Lemma 3.1 on the translated interval of length $C - a$ gives the exact formula

$$\det(m_{i+j}^{(a)})_{0 \leq i, j \leq n} = \frac{(C - a)^{(n+1)^2}}{\prod_{k=0}^n (2k+1) \binom{2k}{k}^2}. \quad (6.1)$$

Its n^2 -root limit is $(C - a)/4$, rather than $C/4$. Indeed, the same lower and upper monic-norm comparisons used above prove this limit directly.

This is a counterexample to a tempting inference about Hankel determinants, not to Kotesovec's conjecture. It explains why a single-entry asymptotic formula cannot safely be substituted into a growing determinant. In the Apéry case, incorrectly discarding the short interval $(0, c)$ would likewise suggest the wrong constant

$$\frac{C - c}{4} = 6\sqrt{2} = 8.485281374\dots, \quad \text{instead of} \quad \frac{C}{4} = 8.492640687\dots$$

The full positive support and the global comparison in Lemma 2.1 retain this small but consequential interval.

7 Exact computations and independent checks

The calculations supplement, but are not used as premises of, the proof. Moments were generated using the classical recurrence

$$(k+1)^3 A_{k+1} = (2k+1)(17k^2 + 17k + 5)A_k - k^3 A_{k-1}, \quad A_0 = 1, A_1 = 5, \quad (7.1)$$

recorded in Edgar's equation (1) and OEIS A005259 [3, 2]. A symmetric fraction-free elimination pass on the 201×201 Hankel matrix produced $D_0, \dots, D_{200}$ as exact integers. At pivot k , the update was

$$a_{ij}^{\text{new}} = \frac{a_{kk}a_{ij} - a_{ik}a_{kj}}{p}, \quad i, j > k, \quad (7.2)$$

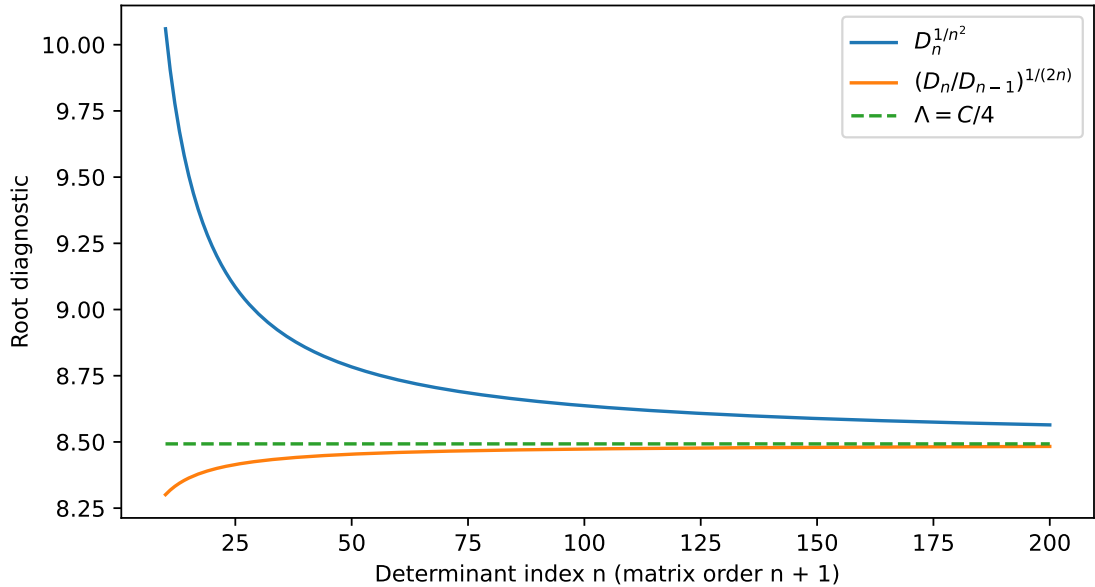
where p is the preceding pivot, initially 1. Every division was checked for zero remainder. Positivity of all leading minors means no pivoting is needed in this application. Successive pivots give the successive leading determinants. There were exactly $200 \cdot 201 \cdot 202/6 = 1,353,400$ checked divisions in the symmetric update; D_{200} has 37,308 decimal digits. The recorded GMP determinant phase took about 124 seconds in this runtime, a measurement rather than a performance guarantee.

Table 2 and Figure 1 use logarithms of these exact integers, not floating-point determinants. Hankel determinant formulas are known to be ill-conditioned in direct numerical use; DLMF §18.2(ix) explicitly warns about this issue [4]. The displayed decimal values are numerical evaluations, not interval-certified bounds. The exact data and exact-arithmetic checks are provided separately.

Check	Count	Result
Recurrence vs. defining binomial sum ($0 \leq k \leq 400$)	401	Pass
Displayed OEIS prefix ($D_0, \dots, D_9$)	10	Pass
Python vs. GMP ($D_0, \dots, D_{40}$)	41	Pass
Independent rational Gaussian determinants ($n \leq 12$)	13	Pass
Independent modular determinants (11 sizes, 3 primes)	33	Pass
Shifted Desnanot–Jacobi identities	64	Pass
Power-weight Cauchy determinants, rational parameters	32	Pass
Central-binomial inequality ($0 \leq k \leq 500$)	501	Pass

Table 1: Exact verification results. The full integer divisions were also checked during generation.

n	D_n^{1/n^2}	$(D_n/D_{n-1})^{1/(2n)}$
1	48.0000000000	6.9282032303
2	20.0577493937	7.6202968816
5	11.9387651922	8.1187083186
10	10.0601045949	8.3009844581
20	9.2400493095	8.3954447392
50	8.7832622606	8.4538328708
100	8.6366837149	8.4732336106
200	8.5643598268	8.4829390793

 Table 2: Root diagnostics computed from exact integers; the common limit is $\Lambda = 8.492640687119 \dots$

 Figure 1: Two root diagnostics from exact determinants, for $10 \leq n \leq 200$. The horizontal line is the proved limit Λ . The finite-sample placement of the curves above or below it is an observation, not a claim of monotonicity or a separate theorem.

The shift experiment file records exact samples at $n = 10, 20, 40, 80$. The three families have $(m, r_n) = (1, n), (2, 0)$, and $(2, n)$. For each sample the shift is fixed throughout that matrix. Samples through $n = 20$ were also computed with the Python implementation. Their numerical roots, compared to the limits in Theorem 4.1, appear in Table 3.

(m, ρ)	Proved limit	$n = 20$	$n = 40$	$n = 80$
$(1, 1)$	131.6072514	154.1754400	142.5207807	136.9744192
$(2, 0)$	288.4997834	360.3947981	322.3570941	304.9315450
$(2, 1)$	6075.4685812	8406.8961687	7151.9818542	6593.0434765

Table 3: Numerical values of $(H_n^{(m, r_n)})^{1/n^2}$ for $r_n = \rho n$. The limits follow from the theorem, not from extrapolation.

8 What has and has not been established

The established conclusion is a proof of the stated OEIS growth limit, using a previously known positive density theorem, with a stronger $O(n)$ error in the logarithm. The same method yields the explicit linear-shift law and the non-D-finiteness corollary. In particular, the work is not only a search for a numerical counterexample: the target conjecture turns out to admit a positive proof.

The analysis does not evaluate the best lower-envelope constant η , determine a limit for h_n/Λ^{2n} , or give a full multiplicative asymptotic equivalent such as $D_n \sim K\Lambda^{n^2}B^n n^\gamma$. None of the other arithmetic assertions on A228143 is being claimed here as a newly solved problem. The literature check establishes the status of the specific accessed entry, not worldwide priority for every consequence of Edgar's theorem.

A natural continuation is to determine the finer normalized norm and determinant asymptotics while treating the interior singular point and the logarithmic endpoint accurately. The elementary comparisons already settle the leading growth questions; finer constants require additional arguments, not extrapolation from the data.

A Reproducing and auditing the package

The supplied files have separate roles. The article contains the proof. The file `data/determinants_0_200.txt` is not distributed with the archive; `build/apery_gmp200` rebuilds it. It contains exact integers; `code/apery_exact.py` is a standard-library Python implementation; `code/apery_gmp.cpp` is the faster C++17/GMP implementation; and `code/verify.py` performs the independent arithmetic checks. The plotting script additionally uses `mpmath` and `matplotlib`. The saved verification log records the actual checks performed.

From the package directory, the exact checks can be rerun with

```
python3 code/verify.py
```

A smaller exact table can be regenerated without external Python packages:

```
python3 code/apery_exact.py -n 40 -output data/recomputed.txt
```

For the full table, compile the supplied GMP implementation and run it with argument 200; the complete commands are in `README.md`. That program reports a failure rather than silently accepting

a nonexact division or nonpositive pivot. `shift_experiments.py` similarly runs shifted examples and can use either implementation.

The symmetric elimination uses $O(n^3)$ integer arithmetic operations and $O(n^2)$ integer storage locations. These are *not* bit-complexity bounds: the integers grow with the determinant size. No claim of a faster asymptotic determinant algorithm is made. The PDF was compiled from this source and rendered for layout inspection. There is no Lean proof or external OEIS submission in the package.

References

- [1] The OEIS Foundation Inc., *A228143: Hankel-type determinants of the Apéry numbers*. Sequence by Zhi-Wei Sun; growth conjecture attributed to Vaclav Kotesovec, 13 September 2026. Internal revision #40, 13 September 2026, accessed 18 September 2026. <https://oeis.org/A228143/internal>.
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- [3] G. A. Edgar, *The Apéry Numbers as a Stieltjes Moment Sequence*, arXiv:2005.10733v2, 2 September 2020. Theorem 1, Proposition 4, Notation 6, Propositions 25–26, Corollary 27, and equation (1). <https://arxiv.org/abs/2005.10733>.
- [4] NIST Digital Library of Mathematical Functions, §18.2, *General Orthogonal Polynomials*, especially the monic norm and Hankel determinant identities and the numerical-conditioning caution in §18.2(ix). Accessed 18 September 2026. <https://dlmf.nist.gov/18.2>.